

VIT BHOPAL UNIVERSITY

PROJECT REPORT

MULTI-SHAPE AREA CALCULATOR USING PYTHON

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Course: ***Introduction to Problem Solving and Programming***

1. Introduction

The Area Calculator is a simple Python program made to calculate the area of different geometric shapes. The program gives the user a menu from which a shape can be selected. After entering the required dimensions, the program calculates and displays the area.

I made this project to practise basic Python concepts such as taking input, using variables, applying conditions, performing calculations and using loops.

2. Project Objective

The main objective of this project is to use basic Python programming concepts to create a small menu-driven application that can calculate the areas of different geometric shapes.

The project also helps in understanding how user input can be processed to produce a useful result.

3. Features

- Simple menu-driven program
- Calculates the area of 8 different shapes
- Takes dimensions from the user
- Displays the calculated area
- Allows multiple calculations in one run
- Displays an invalid-choice message for an unavailable option

4. Shapes and Formulas

Shapes	Formula
1. Circle	$3.14 \times r^2$
2. Rectangle	$\text{length} \times \text{breadth}$
3. Square	side^2
4. Parallelogram	$\text{base} \times \text{height}$
5. Triangle	$0.5 \times \text{base} \times \text{height}$
6. Rhombus	$0.5 \times \text{diagonal 1} \times \text{diagonal 2}$

7. Trapezium
8. Ellipse
axis

$$0.5 \times (a + b) \times \text{height}$$

$$3.14 \times \text{semi-major axis} \times \text{semi-minor axis}$$

5. Technologies and Concepts Used

Python

Variables

input() and print()

if-elif-else statements

while loop

Arithmetic operators

int and float data types

6. Working of the Project

When the program starts, it displays the list of eight shapes. The user enters the number corresponding to the required shape.

The program then asks for the dimensions required for that particular shape. It applies the appropriate formula and displays the calculated area.

After the calculation, the user is asked whether they want to calculate the area of another shape. If they enter Yes, the menu appears again. If they enter No, the program displays a thank-you message.

7. Algorithm

1. Start the program.
2. Set the loop variable to "Yes".
3. Display the menu of available shapes.
4. Ask the user to enter a choice.
5. Take the required dimensions.
6. Apply the appropriate area formula.
7. Display the calculated area.
8. Ask the user whether they want another calculation.
9. If the user enters Yes, repeat the process.
10. If the user enters No/no, end the program.

8. Source Code

Starting the calculator

k="Yes"

while k=="Yes":

Showing all available shapes

print("Area Calculator")

```
print("1. Circle")
print("2. Rectangle")
print("3. Sqare")
print("4. Parallelogram")
print("5. Triangle")
print("6. Rhombus")
print("7. Trapezium")
print("8. Ellipse")

# Selecting a shape
choice=int(input("Enter your Choice:"))

#Circle Calculation
if choice ==1:
    j=float(input("Enter Radius"))
    area=3.14*j*j
    print("Area of Circle=",area)

#Rectangle Calculation
elif choice==2:
    l=float(input("Enter length:"))
    b=float(input("Enter breadth:"))
    area=l*b
    print("Area of Rectangle =",area)

#Square Calculation
elif choice ==3:
    s=float(input("Enter side:"))
    area=s*s
    print("Area of Square=",area)

#Parallelogram Calculation
```

elif choice==4:

b=float(input("Enter base:"))

h=float(input("Enter height:"))

area=b*h

print("Area of Parallelogram=",area)

#Triangle Calculation

elif choice ==5:

b=float(input("Enter base:"))

h=float(input("Enter height:"))

area= 0.5*b*h

print("Area of triangle=",area)

#Rhombus Calculation

elif choice==6:

a=float(input("Enter first diagonol:"))

b=float(input("Enter second Diagnol:"))

area=0.5*a*b

print("Area of Rhombus=",area)

#Trapezium Calculation

elif choice==7:

a=float(input("Enter first parallel side:"))

b=float(input("Enter second parallel side:"))

h=float(input("Enter height:"))

area= 0.5*(a+b)*h

print("Area of Trapezium =",area)

#Ellipse Calculation

elif choice==8:

a=float(input("Enter semi-major axis:"))

b=float(input("Enter semi-minor axis:"))

```

    area=3.14*a*b
    print("Area of Ellipse")

# If the entered option is not available
else:
    print("Invalid Choice")

# Asking if the user wants to continue
k=input("Do you want know the area of more shapes(Type Yes/No)")

# Closing the program
if k=="No" or "no":
    print("Thank you!!!")

```

9. Testing

Test Case	Input	Expected Result
1. Circle	Radius = 5	Area = 78.5
2. Rectangle	Length = 10, Breadth = 5	Area = 50
3. Square	Side = 6	Area = 36
4. Triangle	Base = 8, Height = 5	Area = 20

5. Rhombus

Diagonals = 10, 6

Area = 30

6. Invalid choice

Choice = 9

Invalid Choice

The program should also be tested by selecting different shapes and checking whether the displayed result matches the corresponding formula.

10. Result

- The program provides a simple way to calculate the area of different geometric shapes. It takes the required measurements from the user, performs the calculation and displays the result.

The use of a while loop also allows the user to perform multiple calculations without restarting the program.

11. Conclusion

- This project helped me understand how different basic Python concepts can be combined to make a working program. While making it, I practised taking user input, using conditional statements, performing calculations and repeating a task using a loop.

It also helped me understand how mathematical formulas can be converted into Python statements and used in a practical program.

12. Future Improvements

In the future, the project can be improved by:

- Adding input validation for incorrect or negative values.
- Adding more geometric shapes.
- Using functions to make the program shorter and easier to manage.
- Adding a graphical user interface.

13. Project Links

<https://github.com/Jagriti-Talwar/Area-Calculator-Python>

OneCompiler: <https://onecompiler.com/python/453tbzpke>

GitHub Repository:

<https://github.com/Jagriti-Talwar/Area-Calculator-Python>

14. Screenshots

```
PS C:\Users\Jagriti Talwar\OneDrive\Desktop\Area-Calculator-Python> python area_calculator.py
Area Calculator
1. Circle
2. Rectangle
3. Square
4. Parallelogram
5. Triangle
6. Rhombus
7. Trapezium
8. Ellipse
Enter your Choice:1
```

```
4. Parallelogram
5. Triangle
6. Rhombus
7. Trapezium
8. Ellipse
Enter your Choice:1
Enter Radius:7
Area of Circle= 153.86
Do you want know the area of more shapes(Type Yes/No):no
Thank you!!!
PS C:\Users\Jagriti Talwar\OneDrive\Desktop\Area-Calculator-Python>
```

```
1. Square
2. Parallelogram
3. Triangle
4. Rhombus
5. Trapezium
6. Ellipse
Enter your Choice:4
Enter base:7
Enter height:8
Area of Parallelogram= 56.0
Do you want know the area of more shapes(Type Yes/No):ye
```

```
PS C:\Users\Jagriti Talwar\OneDrive\Desktop\Area-Calculator-Python> python area_calculator.py
Area Calculator
1. Circle
2. Rectangle
3. Square
4. Parallelogram
5. Triangle
6. Rhombus
7. Trapezium
8. Ellipse
Enter your Choice:4
```

```
PS C:\Users\Jagriti Talwar\OneDrive\Desktop\Area-Calculator-Python> python area_calculator.py
Area Calculator
1. Circle
2. Rectangle
3. Square
4. Parallelogram
5. Triangle
6. Rhombus
7. Trapezium
8. Ellipse
Enter your Choice:5
```

```
5. Triangle
6. Rhombus
7. Trapezium
8. Ellipse
Enter your Choice:5
Enter base:9
Enter height:7
Area of triangle= 31.5
Do you want know the area of more shapes(Type Yes/No):no
Thank you!!!
PS C:\Users\Jagriti Talwar\OneDrive\Desktop\Area-Calculator-Python>
```

